



Indian Institute of Technology Guwahati

BSc. (Hons.) Data Science & Artificial Intelligence
Online Degree Programme

INTERNSHIP/TERM PROJECT

Ultrasonic Weld Quality Analytics Platform Using Machine Learning and Digital Image Processing

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Trimester :	VIII

PROJECT BACKGROUND

- **Introduction**

Ultrasonic welding is a **solid-state joining process** widely used in the **automotive, electronics, medical, and packaging industries**. Reliable inspection is essential to ensure **product quality, structural integrity, and manufacturing consistency**.

- **Problem Statement**

- Manual ultrasonic weld inspection is **time-consuming**.
- Inspection quality depends on **operator expertise**.
- Small weld defects are difficult to detect consistently.
- Manual inspection limits productivity in high-volume manufacturing.

- **Motivation**

Develop an **AI-assisted Ultrasonic Weld Quality Analytics Platform** that:

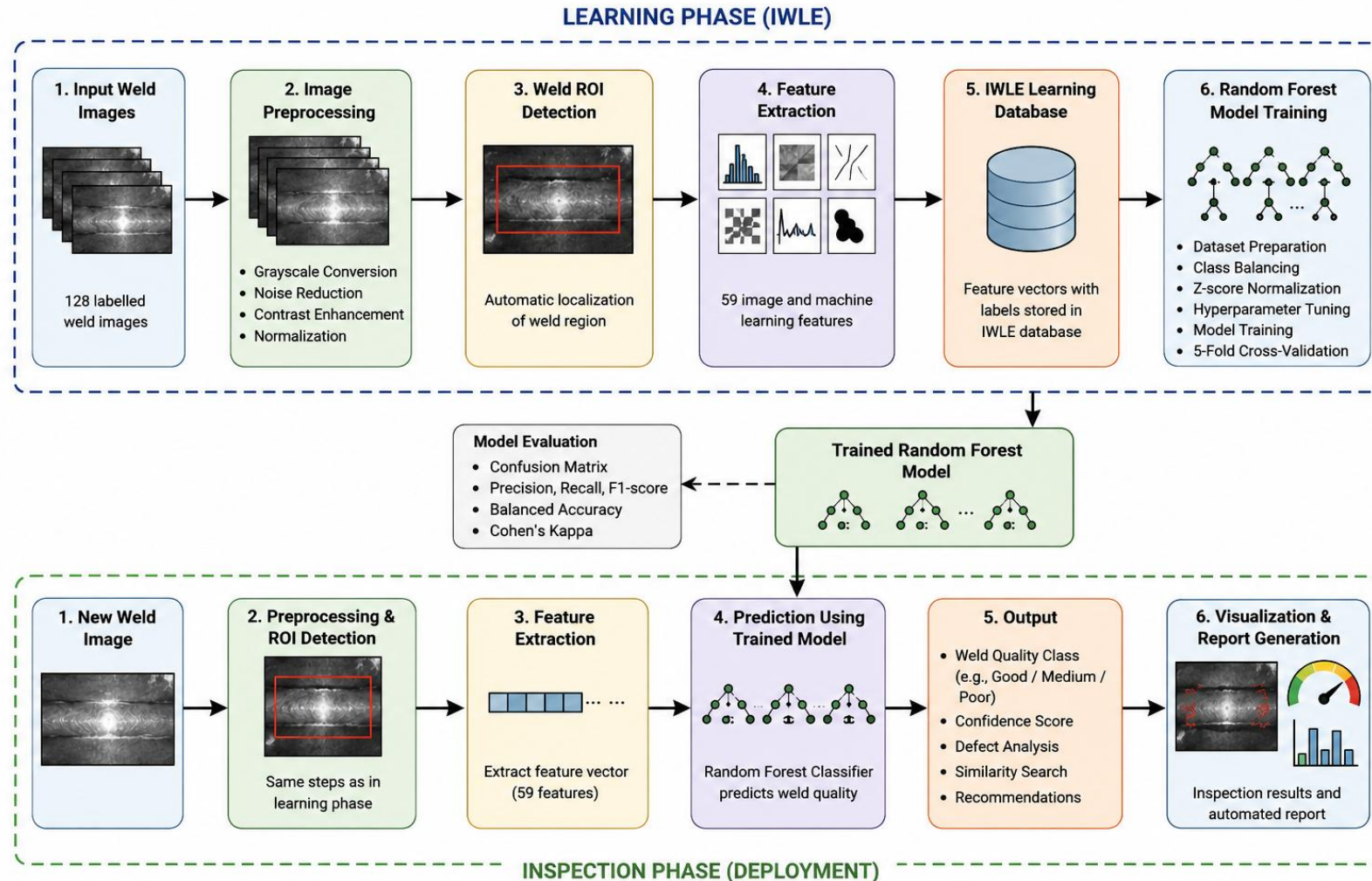
- Automates ultrasonic weld image analysis.
- Detects defects using **Digital Image Processing**.
- Predicts weld quality with a **Random Forest** model.
- Generates objective quality assessments and engineering recommendations.

PROJECT OBJECTIVES

- **Objectives**

- Develop an **AI-assisted Ultrasonic Weld Quality Analytics Platform** for automated weld inspection.
- Apply **Digital Image Processing** techniques to enhance and analyze ultrasonic weld images.
- Build a **Random Forest** machine learning model to classify weld quality.
- Detect weld defects and generate objective quality assessments.
- Provide engineering recommendations through an interactive **MATLAB GUI**.
- Improve inspection **accuracy, consistency, and efficiency** while reducing manual effort.

SYSTEM ARCHITECTURE & WORKFLOW



TOOLS & TECHNOLOGIES

- **Software & Development Tools**

- **MATLAB R2025b** – Application development and implementation
- **Image Processing Toolbox** – Image enhancement, ROI detection, and feature extraction
- **Statistics and Machine Learning Toolbox** – Random Forest model development
- **MATLAB App Designer** – Interactive graphical user interface (GUI)

- **Core Technologies**

- **Digital Image Processing**

- Image preprocessing
- ROI detection
- Feature extraction

- **Machine Learning**

- Random Forest classifier
- Model training and prediction
- Performance evaluation

- **Dataset**

- **128 labelled ultrasonic weld images**
- Used for model training, validation, and testing

METHODOLOGY

- **Image Processing**

- Image acquisition and grayscale conversion
- Noise reduction and contrast enhancement
- Automatic weld **Region of Interest (ROI)** detection
- Extraction of **59 image features**

- **Machine Learning**

- Feature normalization
- Random Forest classifier training
- Hyperparameter tuning
- 5-fold cross-validation
- Weld quality prediction

- **System Output**

- Weld quality classification
- Confidence score
- Defect analysis
- Engineering recommendations
- Automated report generation

Graphical User Interface (GUI)

Ultrasonic Weld Quality Analytics Platform

ULTRASONIC WELD QUALITY ANALYTICS PLATFORM

Inspection & Defect Analysis Dashboard

CONTROL CENTER

IMAGE INPUT

Load Image Camera

Batch Folder Select ...

ANALYSIS CONTROLS

Analyze Clear

Export PDF Export Excel

WELD PARAMETERS

Material Type	Copper
Thickness (mm)	0.5
Weld Time (s)	0.35
Pressure (MPa)	0.25
Amplitude (%)	85
Frequency (kHz)	20

IMAGE ANALYSIS WORKSPACE

ORIGINAL IMAGE

ENHANCED IMAGE

DEFECT VISUALIZATION

Overlay

CRACK SEGMENTATION

QUALITY DASHBOARD

QUALITY SCORE 87%	CONFIDENCE 71%
CRACK COUNT 0	SEVERITY LOW
STATUS GOOD	

AI RECOMMENDATIONS

```
-----
AI WELD ASSESSMENT REPORT
-----
Inspection Time : 29-Jul-2026 00:56:41
Material       : Copper
Thickness      : 0.5 mm

-----
1. MACHINE LEARNING ASSESSMENT
-----
Prediction     : GOOD
Confidence      : 71.3 %

-----
2. IMAGE ANALYSIS
-----
```

ANALYSIS METRICS

	Metric	Value
1	Material	Copper
2	Thickness	0.5 mm
3	Parameter Score	100 %
4	Current Frequency	20 kHz
5	Recommended Frequency	20 kHz
6	Quality Score	86.51
7	Confidence	79.61 %
8	Crack Count	

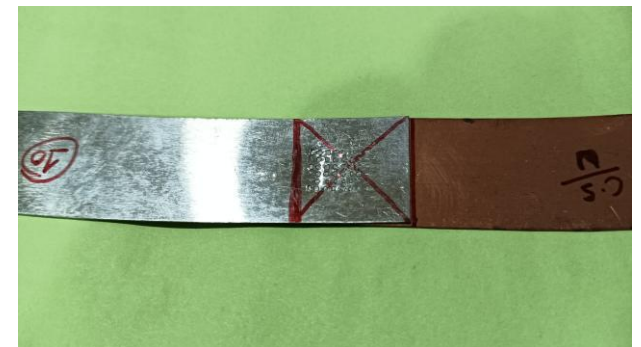
EXPERIMENTAL RESULTS

Parameter	Result
Material	Copper
Thickness	0.5 mm
Quality Score	86.51%
Rule-Based Inspection Confidence	79.61%
AI Prediction	GOOD
AI Confidence	71.3%
Defect Type	NO DEFECT
Crack Count	0
Severity	LOW
Overall Decision	GOOD

- **Key Observations**

- No significant weld defects were detected.
- Random Forest prediction agreed with the rule-based inspection.
- Parameter compatibility reached **100%**.
- Frequency matched the recommended value (**20 kHz**).
- The system concluded that **no corrective action was required**.

QUALITY DASHBOARD	
QUALITY SCORE	CONFIDENCE
87%	71%
CRACK COUNT	SEVERITY
0	LOW
STATUS	
GOOD	



Conclusion & Future Work

Conclusion

- Developed an **AI-assisted Ultrasonic Weld Quality Analytics Platform** using MATLAB.
- Successfully integrated **Digital Image Processing** with a **Random Forest** machine learning model.
- Automated ultrasonic weld inspection, defect detection, and quality assessment.
- Generated engineering recommendations and automated inspection reports through an interactive GUI.
- Demonstrated reliable and efficient ultrasonic weld quality evaluation with objective decision support.

Future Scope

- Expand the dataset to improve model robustness and generalization.
- Investigate advanced machine learning and deep learning techniques.
- Enable real-time ultrasonic weld inspection for industrial applications.
- Integrate with Industrial IoT (IIoT) and smart manufacturing systems.
- Develop a cloud or edge-based deployment for scalable inspection.

Thank You